

IMAGE CAPTIONING AND DIFFUSION

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MASTER IN
COMPUTER VISION
Barcelona



GitHub Repository

https://github.com/JordiVentosa/C5_Team3

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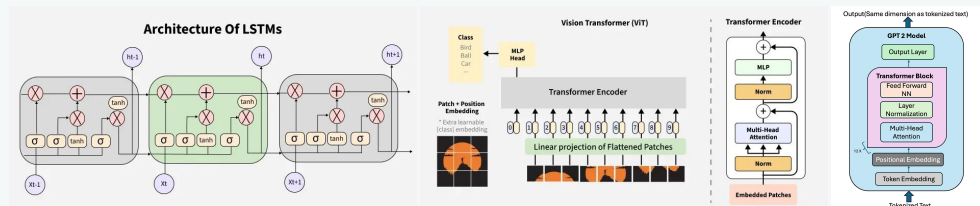
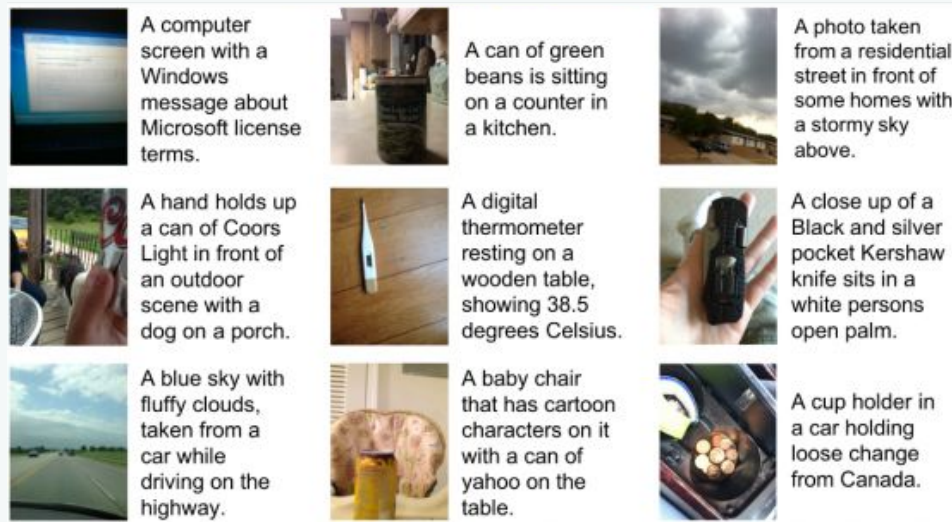
1. Introduction

1.1 Image Captioning in VizWiz Dataset

Objective: Give a synthesized description of the content of an image for people with vision problems.

Priorities: We want to improve the BLEU-1, BLEU-2, ROUGE-L and METEOR metrics, but consistently. As we're going to see later, not always the improvement of this metrics reflects in a better performance.

Difficulties: Small details on the images, hard pictures with complex photographic issues, poor labeling in many examples, among others.



1.2 Dataset

We will use the **VizWiz dataset** with the following train/val/test configuration:

Split	Images	Captions	Origin
Train	21,087	105,435	90% train folder
Validation	2,344	11,720	10% train folder
Test	7,750	38,750	100% val folder
TOTAL	31,181	155,905	

Depending on the tokenizer, the size of the vocabulary and the maxim length will be different.

Tokenizer Type	Vocab Size	Max Sequence Length
Character	80	641
Word	8,596	115
Subword	30,522	156

The dataset contains thousands of images with a ratio of **5 captions per image**. The captions have an average length of **11.64 words** with a median at 10, with 99% of the captions contain **27 words** or fewer. The max sequence length and vocabulary sizes showed in the table are extracted from the training set, and as we can see because the max length for Word is 115, we have a small outliers that elevate the max sequence length. We kept that length, but if we had to train a significantly bigger model, it could have been a good idea to eliminate these outliers in order to speed up the training.

2. Captioning with classic DL

2.1 Experimental method

Because of the limited resources and time, perform an extensive hyperparameter search of all the combinations is not viable, for this reason we adopted a progressive methodology in order to acute the architecture.

We start with the baseline hyperparameters and perform small experiments to update the Baseline across them, finishing with a full hyperparameter search for training

Initial Baseline

Encoder: Resnet-18
Decoder: GRU
Tokenizer: Character
Optimizer: AdamW
Learning rate: 1e-4
Batch size: 256
Scheduler: None
TeacherF ratio: No

1. Encoder

Resnet-18
vs.
Resnet-34

2. Decoder

GRU
vs.
LSTM

3. Token Level

Character,
subword and word

4. Attention

Try Bahdanau
attention

5. Full Search

Bayesian search for LR,
Teacher Forcing (TF),
optimizers and schedulers

2.2 Ablation Study: Observed Behaviour

1. Encoder

Resnet-18
vs.
Resnet-34

→ Big improvement especially in BLEU metrics

→ The model produces longer captions, but still nonsense

2. Decoder

GRU
vs.
LSTM

→ LSTM improves BLEU-1, but worsen the other metrics.

→ LSTM seems to produce a lot of empty spaces and fill-in tokens

3. Token Level

Character, subword
and word

→ Word gives the best results, followed by subword

→ LSTM here improves when subword is used, but GRU with word tokens win

4. Attention

Try Bahdanau
attention

→ Tried Bahdanau Attention between encoder and decoder

→ The metrics slightly worsen, although probably insignificant at a statistical level

5. Bayesian Search

LR, TF, optimizers
and schedulers

→ Performed a hyperparameter search in LR, Teacher Forcing ratio, optimizers and scheduler

→ No effect, practically same metrics

2.3 Quantitative Results

If we take a final look at the quantitative results, now with the test set values, we see that the optimization of the hyperparameters did nothing, since the test metrics are the same or even slightly worse. Indeed, we can see a clear ceiling on the capabilities of this architecture that it seems we've reached. In conclusion, some experiments helped to improve our pipeline, and we discovered the maximum available capabilities of the best architecture on this dataset.

Validation (highlighted experiments)

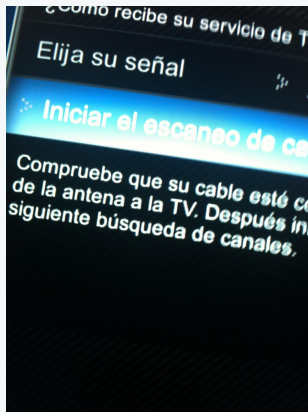
	Metric vs. Model	BLEU-1	BLEU-2	ROUGE-L	METEOR
●	Initial Baseline	0.056	0.039	0.172	0.11
●	Modified Encoder (ResNet34)	0.098	0.073	0.185	0.133
	Modified Decoder (LSTM)	0.158	0.043	0.126	0.096
●	Modified Token Level (GRU Word)	0.207	0.139	0.294	0.202
	Added Attention (Bahdanau)	0.199	0.131	0.285	0.191
●	Optimized Hyperparameters	0.210	0.143	0.290	0.204

Test (models improvements)

	Metric vs. Model	BLEU-1	BLEU-2	ROUGE-L	METEOR
●	Initial Baseline	0.059	0.043	0.177	0.114
●	Modified Encoder (ResNet34)	0.061	0.044	0.173	0.118
●	Modified Token Level (GRU Word)	0.201	0.136	0.286	0.196
●	Optimized Hyperparameters	0.201	0.135	0.279	0.196

**Only tested after performing all the experiments and on the models that caused an improvement to our baseline*

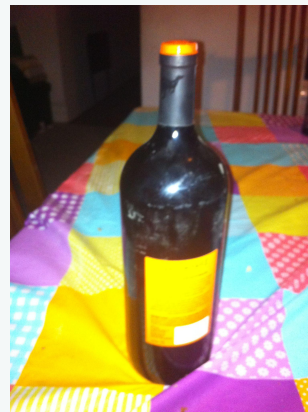
2.4 Qualitative Results



Ini.Base: "A blrte"
 Mod. Enc.: "A coite"
 Mod. Dec.: "A alite se a
 oo o e isual
 content."
 Mod. Tkn.: "a screen screen a
 a a recognize a"
 Mod. Att.: "a computer screen
 a a a a a"
 Mod. Hyp.: "a computer
 screen a a a a a"
 GT: "A settings screen of
 selecting a menu option in
 Spanish text."



Ini.Base: "A per o o"
 Mod. Enc.: "A baa o"
 Mod. Dec.: "A alite se a oo
 o e isual content."
 Mod. Tkn.: "a issues of a a a a a a"
 Mod. Att.: "a person of a a a a a"
 Mod. Hyp.: "a dvd of a a a a a"
 GT: "a ps3 game for MLB 18 the
 show"



Ini.Base: "A blat o"
 Mod. Enc.: "A ality issue ar
 o eeee ni isaa
 onnttt."
 Mod. Dec.: "A alite se a
 oo o e isual
 content."
 Mod. Tkn.: "a bottle bottle a a
 a a a a"
 Mod. Att.: "a bottle of of a a a
 a a"
 Mod. Hyp.: "a person with a a
 a a a"
 GT: "a black bottle with yellow
 label placed on a fabric
 materials"



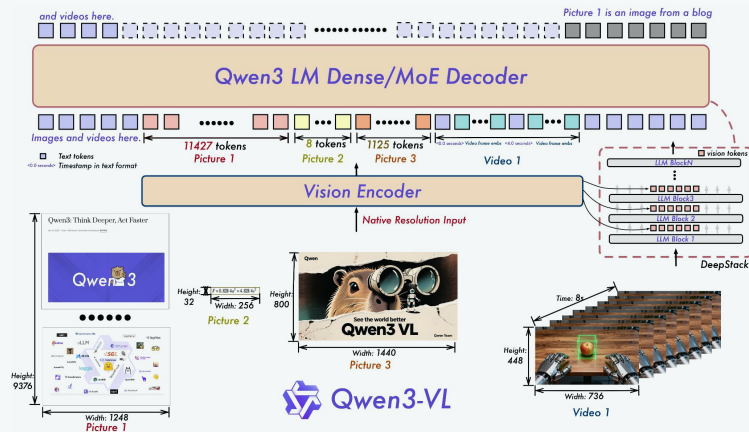
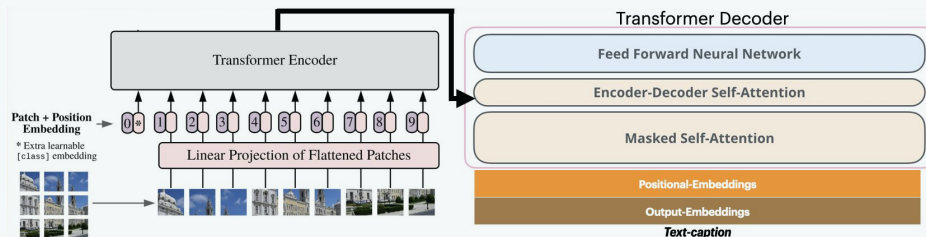
Ini.Base: "A bac"
 Mod. Enc.: "A cax o"
 Mod. Dec.: "A alite se a
 oo o e isual
 content."
 Mod. Tkn.: "a box of of a of a
 a"
 Mod. Att.: "a box of of of on
 on"
 Mod. Hyp.: "a of of of of on
 on"
 GT: "A black plastic container
 with salmon inside of it"

3. Captioning with Transformers

3.1 Introduction

After exploring classical encoder-decoder architectures, we wanted to push our results further using more modern approaches:

- Pretrained ViT+GPT2 encoder-decoder model, fine-tuned under different strategies to adapt it to our domain
- Multimodal large language models from the Qwen family, explored in a zero-shot setting and fine-tuned using LoRA



3.2 ViT + GPT2 Inference

- Tested the pretrained ViT+GPT2 directly on our validation set.
- Promising initial metrics, but problems with some predictions.
- We tried other backbones and captioners, but none of them gave comparable results.

	Meteor	Rouge-l	Bleu-1	Bleu-2
Baseline	0.283	0.367	0.542	0.340



Model: A bottle of water sitting on top of a table

Reference: A bottle of germ-x sanitiser that is empty on the counter



Model: A cat laying on the floor next to a table

Reference: Quality issues are too severe to recognize visual content

3.3 ViT + GPT2 Finetuning

- **Three finetuning strategies:** Backbone, Captioner & All
- All of them improved on the original model on the meteor benchmark
- **Best fine tuning:** Captioner
- **Potential Causes:** Insufficient data, lot of burry images

Unfrozen	Meteor	Rouge-l	Bleu-1	Bleu-2
Baseline	0.283	0.367	0.542	0.340
Captioner	0.424	0.429	0.495	0.362
Backbone	0.420	0.356	0.278	0.208
All	0.414	0.357	0.270	0.202



Captioner: Quality issues are to severe to recognize visual content

Backbone: A white wall with a brown and black object on it

All: A white wall with a black cord hanging from it

Reference: Quality issues are to severe to recognize visual content



Captioner: Quality issues are to severe to recognize visual content

Backbone: A close up of a purple fabric with some wrinkles

All: A close up of a piece of fabric that is brown in color

Reference 1: Quality issues are to severe to recognize visual content

Reference 2: A blue piece of material sitting on some sort of surface

3.4 Zero-shot Multimodal LLMs

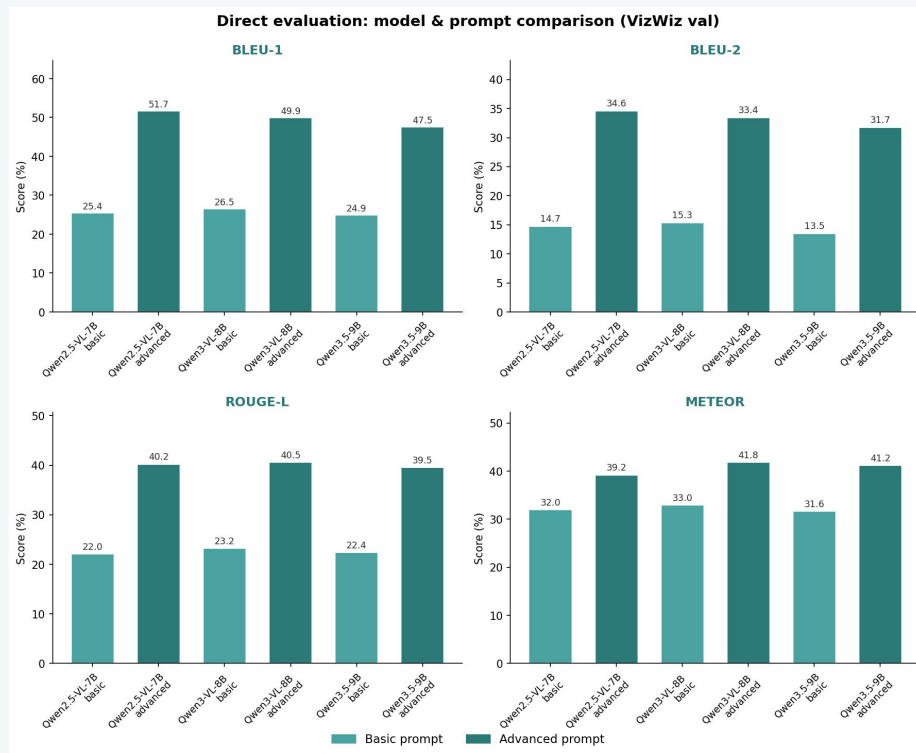
Models evaluated:

- **Qwen2.5-VL-7B** (~14GB)
- **Qwen3-VL-8B** (~16GB)
- **Qwen3.5-9B** (~18GB)

Two prompt strategies for each model:

- **Basic:** "Describe this image briefly"
- **Advanced:** domain-specific system prompt for VizWiz indicating that images may be blurry or poorly framed.

Best results: **Qwen3-VL-8B** with **Advanced prompt**



3.4 Zero-shot Multimodal LLMs



References:

1. a bottle of germ-x hand sanitizer that is empty on the counter.
2. a clear plastic bottle containing germicide gel on a white computer desk with someone's name written in sharpie on the back
3. a bottle of hand sanitizer is almost empty

Qwen2.5-VL-7B - Advanced prompt

a bottle of germ-x hand sanitizer with a handwritten label is placed on a desk near a keyboard and mouse.

Qwen3-VL-8B - Advanced prompt

a clear bottle of germ-x hand sanitizer with "dapacom" handwritten on the label, sitting on a desk next to a keyboard.

Qwen3.5-9B - Advanced prompt

a bottle of germ-x hand sanitizer with a blue cap, sitting on a desk next to a keyboard and other office items.

3.5 ViT + LoRA-adapted Qwen3.5

The architecture combines a frozen **ViT encoder** (*nlpconnect/vit-gpt2-image-captioning*, 86M params) with a **Qwen3.5** text decoder connected via a learned **Projection MLP**.



We experimented with two decoder sizes (0.8B and 4B), different LoRA ranks ($r=8, 16$), scaling factors ($\alpha=16, 32$), learning rates ($5e-4, 1e-4, 5e-5$) and dropout values ($0.010, 0.025, 0.050$). Post-processing also had a significant impact on the final metrics.

	Meteor	Rouge-l	Bleu-1	Bleu-2
Without post-processing	0.355	0.278	0.249	0.170
With post-processing	0.382	0.441	0.572	0.423

3.5 ViT + LoRA-adapted Qwen3.5



References:

1. someone was doing a google and they received a message from swagbucks saying they won.
2. a picture of computer screen that has swag bucks on display.
3. a computer monitor screen showing " congratulations! you've won 11 swag bucks"

Best model: Qwen3.5-4B, $r=8$, $\alpha=16$, $lr=1e-4$, dropout=0.05

Prediction:

"assistant

a computer screen with a website that is asking for a user's name and email address. the website is for a company called "the online store". the website is in a language that is not english. the website is on a computer screen that is blue and white. the website is on a computer"

Best model: Qwen3.5-4B, $r=8$, $\alpha=16$, $lr=1e-4$, dropout=0.05 (with post-processing)

Prediction:

"a computer screen with a website that is asking for an email address."

Prediction with best model Task 2.1:

Prediction:

"a computer screen displaying a swagbucks website claiming the user won 11 swag bucks, with a captcha verification prompt."

4. Improving with Synthetic Data

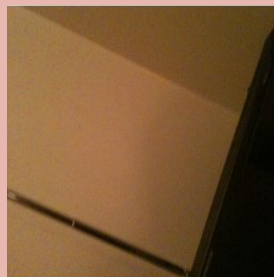
4.1 Dataset problem

The main issue we find in the dataset comes from the number of images that are labeled as “Quality Issues”, so our aim with diffusion models was to try to remove as many of this captions as possible.

Lite Quality Issues



Big Quality Issues

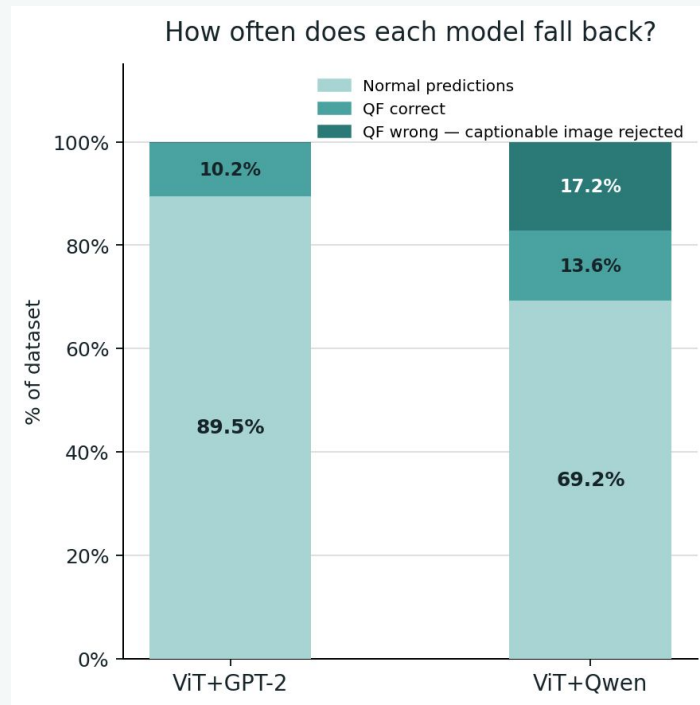


4.1 Dataset problem

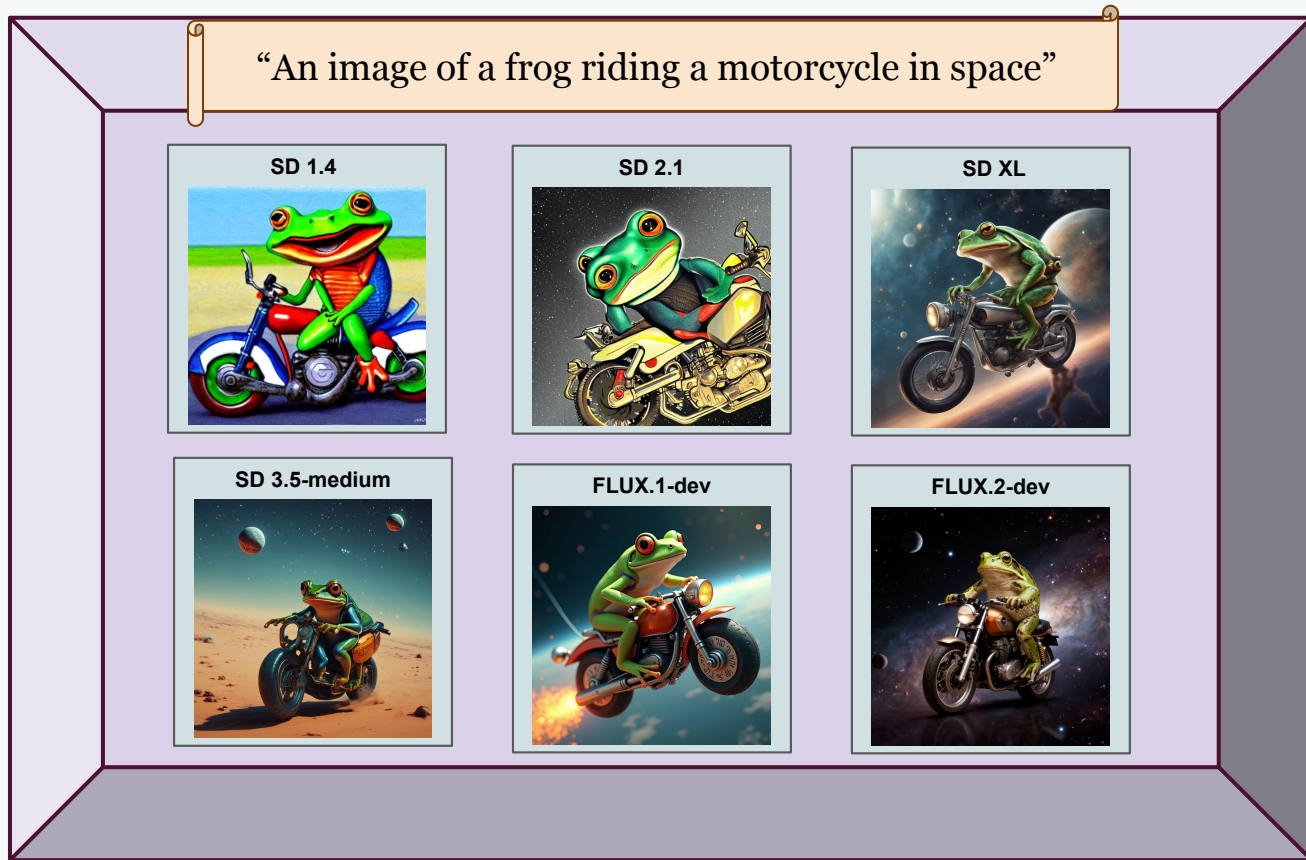
- ViT + Qwen: 30% Total, 17% wrong
- ViT + GPT-2: 10% Total

Even though ViT is technically correct, a lot of images with quality issues also have actual captions, and we want the models to focus on those more

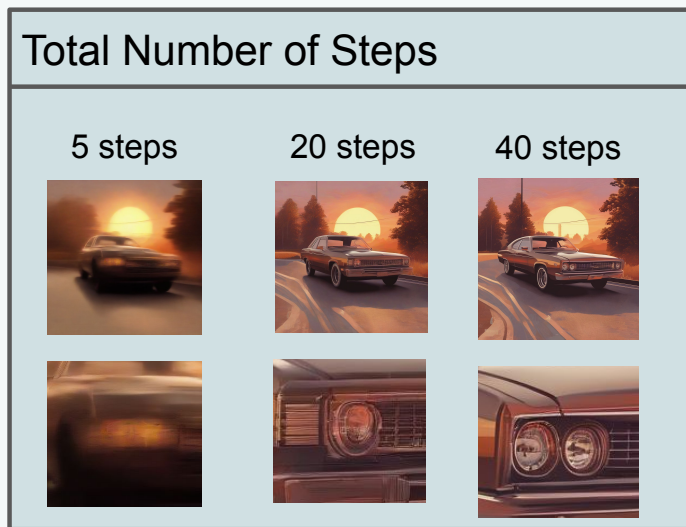
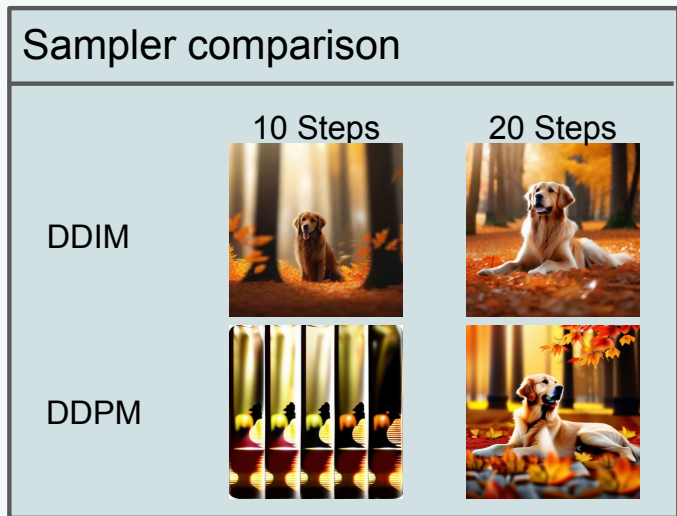
Predicted caption	Reference examples
"quality issues are too severe to recognize visual content"	"a computer screen shows a repair prompt" "partial computer screen showing the need of repairs"



4.2 Qualitative Model Comparison



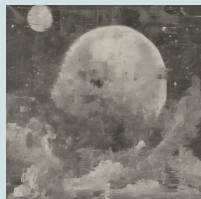
4.3 Model Experiments



4.3 Model Experiments

CFG

1



3

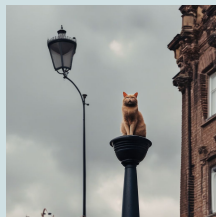


10

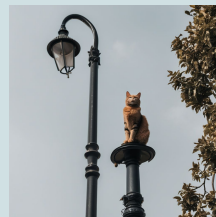


Negative and Positive Prompting

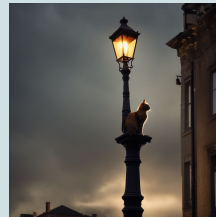
None



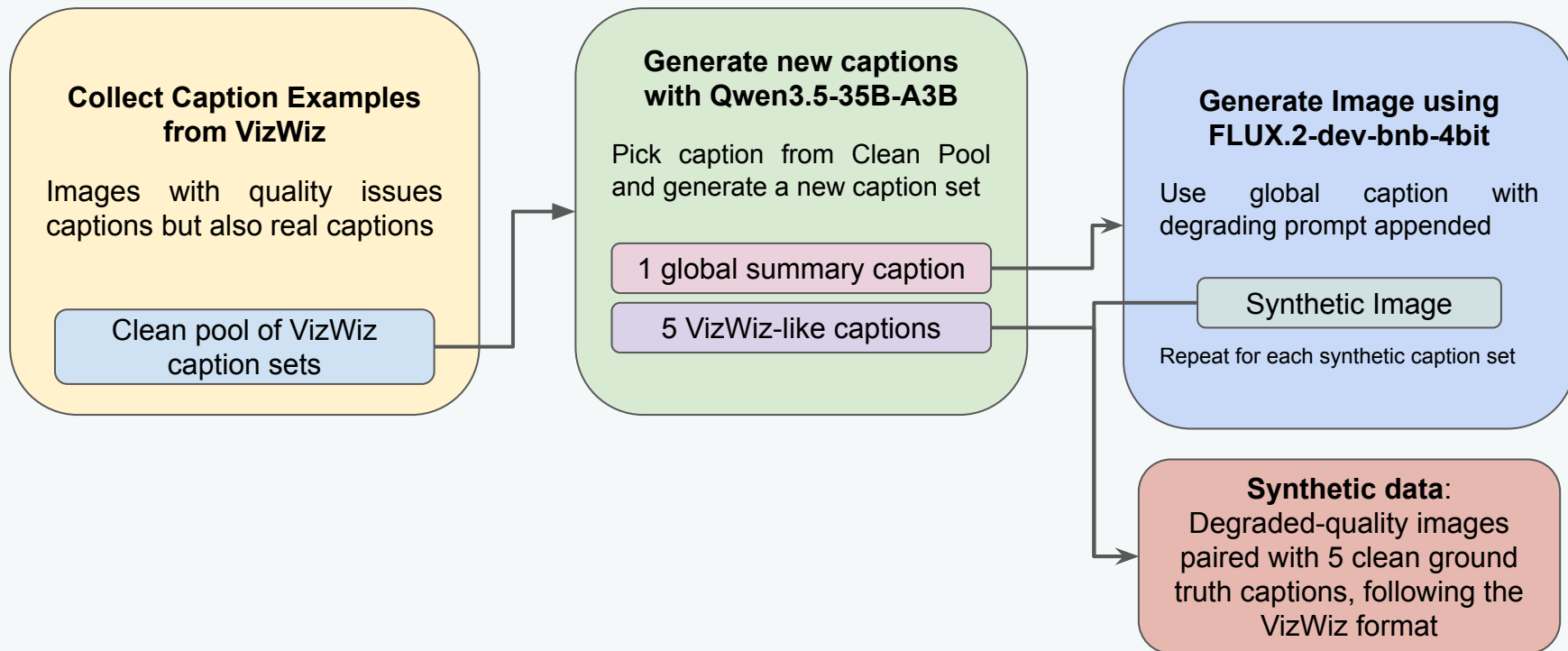
Positive



Negative



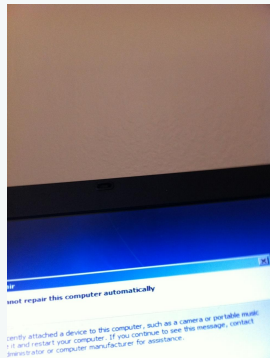
4.4 Synthetic Data Generation Pipeline 1



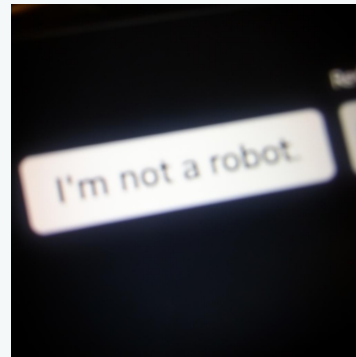
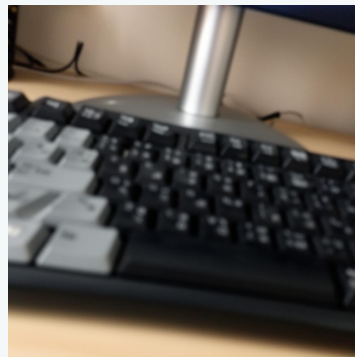
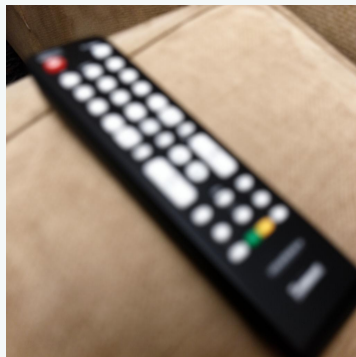
4.4 Synthetic Data Generation Pipeline 1

- **2000** Synthetic images
- **Similar style** to original
- Qwen makes some **mistakes**

Real VizWiz



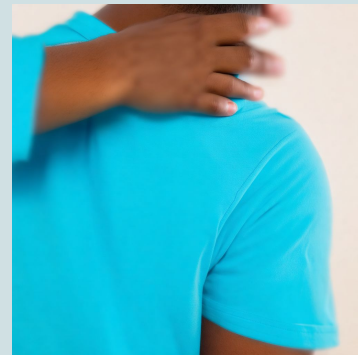
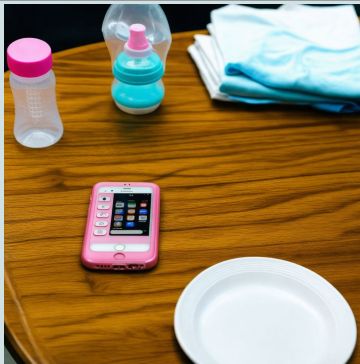
Synthetic
(FLUX.2-dev-bnb-4bit)



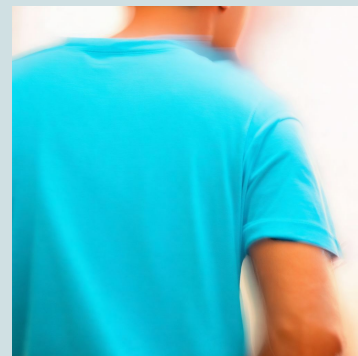
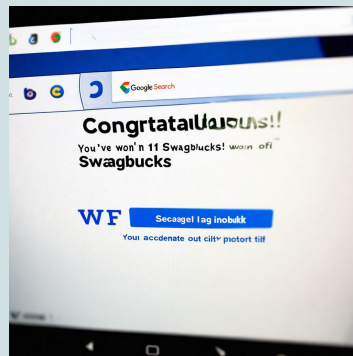
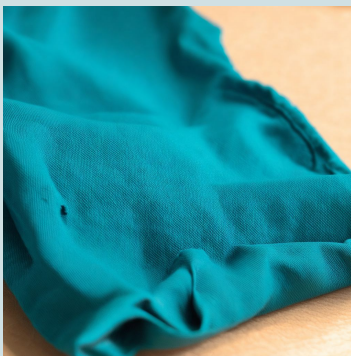
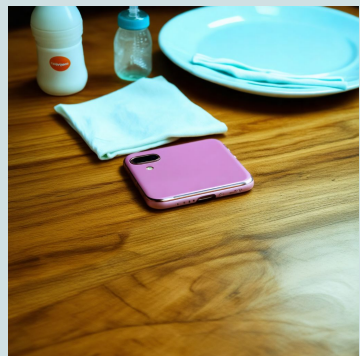
4.5 LoRA Fine-tune Stable Diffusion 3.5 -medium

Stable Diffusion 3.5

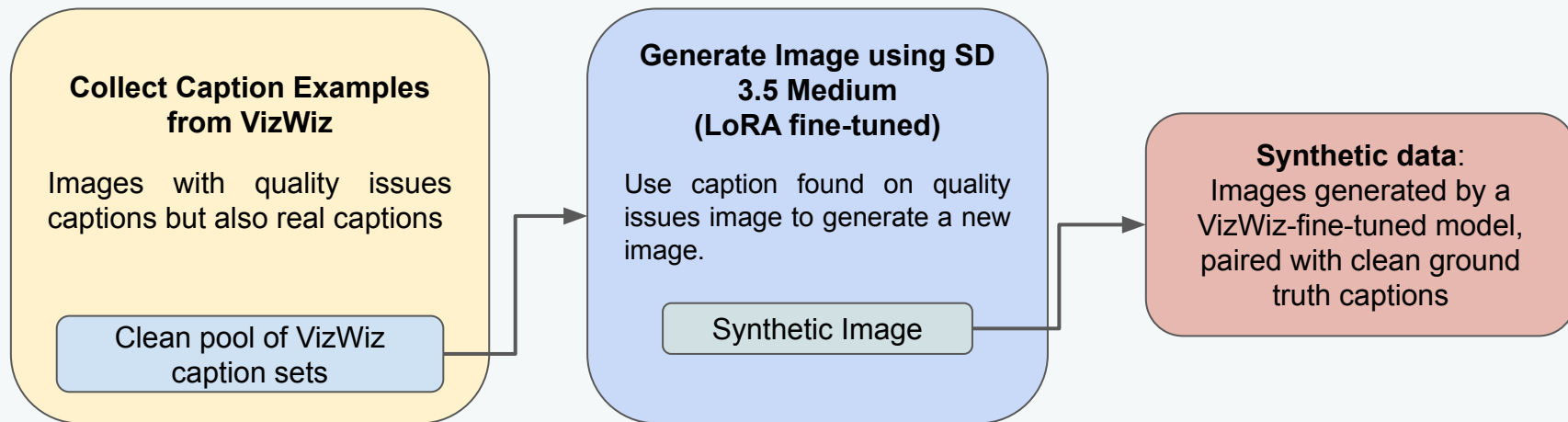
(using same prompts)



**Stable Diffusion 3.5
fine-tuned**

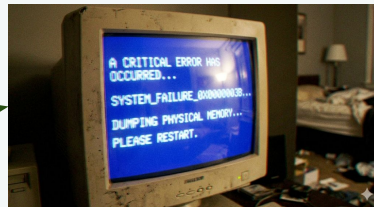


4.6 Synthetic Data Generation Pipeline 2



~~Caption 1: Quality issues are too severe to recognize visual content.~~

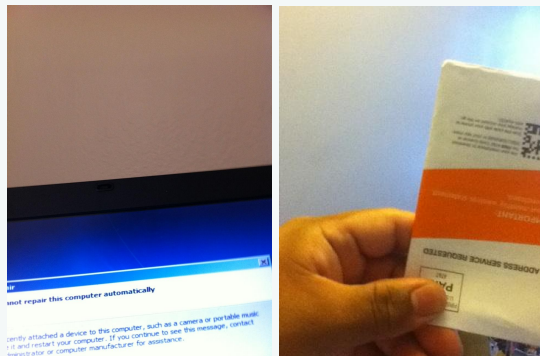
Caption 2: A computer screen with a blue background and white text



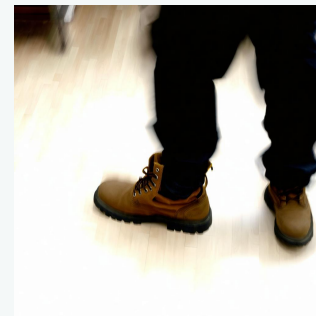
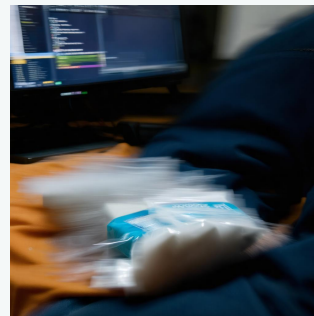
4.6 Synthetic Data Generation Pipeline 2

- **2000** Synthetic images
- **Similar style** to original
- Stable diffusion **hallucinates** more than Flux2

Real VizWiz



Synthetic
(SD 3.5 Medium
fine-tuned)



4.7 Training Strategies

1. Using **only** the **original data**, but **removing quality issue** captions from images that had at least two valid captions
2. Using the **original data combined** with the **FLUX.2** synthetic images
3. Using the **original data combined** with the **SD 3.5 Medium LoRA** synthetic images
4. Using the **original data combined** with **both synthetic sets**

Approach 1

Original Data
(No Quality Issues)

Approach 2

Original Data

Synthetic
FLUX.2

Approach 3

Original Data

Synthetic
SD 3.5 Medium LoRA

Approach 4

Original Data

Synthetic
FLUX.2

Synthetic
SD 3.5 Medium LoRA

4.8 Quantitative Results

Removing **Quality Issues Caption** lowers metrics a lot

Qwen caption style leads to worse metrics but good qualitative

Fine Tuned SD3.5 reduces Quality Issues Captions while maintaining metrics

Approach	Meteor	Rouge-L	Bleu-1	Bleu-2
Best Week 4	0.382	0.441	0.572	0.423
Original Data (No Quality Issues)	0.319	0.256	0.234	0.152
Original Data + Synthetic FLUX.2	0.381	0.417	0.551	0.366
Original Data + Synthetic SD 3.5 Medium LoRA	0.373	0.421	0.609	0.412
Original Data + Synthetic FLUX.2 + Synthetic SD 3.5 Medium LoRA	0.356	0.396	0.571	0.368

4.9 Qualitative results



Reference: Quality issues are too severe to recognize visual content.

Removed captions: A computer screen with a blue background and white text

FLUX-2: A computer screen with a blue background and the windows start menu open

SD-3.5: A computer screen with a blue background and white text

FLUX-2 + SD-3.5: A computer screen with a blue background and the windows start menu open

5. Conclusions

5. Conclusions

Architecture Evolution

- Classic architectures reached a clear performance ceiling on the VizWiz dataset, and the training became too difficult
- Transformers and LLMs delivered massive improvement and allowed domain-specific understanding
- The models fall on shortcuts, inflating the metrics

Diffusion and Synthetic

- Diffusion models can be used for quality synthetic data generation, even in datasets as difficult like VizWiz
- The management of the quality issues fallback caption succeeded in improving the performance
- The metrics don't fully represent the improvement, but now the models are more concise

Approach	Meteor	RougeL	Bleu-1	Bleu-2
Classic DL	0.201	0.136	0.286	0.196
ViT + GPT2 (Fine-tuned All)	0.270	0.202	0.357	0.414
Qwen3.5 + LoRA	0.382	0.441	0.572	0.423
Original + Synthetic SD 3.5 LoRA	0.373	0.421	0.609	0.412

Thank you for your
<https://arxiv.org/abs/1706.03762>